Subject Description Form

Subject Code	ITC6861RT
Subject Title	Guided Study in Computational Methods for Problems in Fashion & Textiles Industry
Credit Value	3
Level	6
Pre-requisite / Co- requisite / Exclusion	Nil
Objectives	This course provides students with solid background in the theoretical and practical application of computational methods in solving a chosen type of problem in the textiles and fashion industry. Students need to select and focus on a particular class of problem. The main objectives are: a. to train up the students to understand computational method; b. to apply the computational methods; c. to evaluate the computational methods; and d. to select the right method and tools to solve the problem. Students are required to attend seminars, read assigned readings, complete a computational project, and an examination.
Intended Learning Outcomes	Students who have successfully completed the subject should be able to: a. Understand the theory of the computational methods; b. Select the right computational methods to solve a problem; c. Apply theory and methods in solving a problem; d. Evaluate the performance and limitations of the computational methods.
Subject Synopsis/ Indicative Syllabus	This subject aims to provide students with theoretical and professional knowledge in advanced computational methods in solving industrial problems. The syllabus is divided into THREE main parts: Part 1: Introduction to computational methods • According to the selected class of problem, a full class of computational methods are introduced; • Introduction to the theoretical analysis on the computation methods; • Introduction to the procedural content of the computation m • Conducting the problem solving. Part 2: Analysis of computational methods • Introduction to the performance evaluation, such as speed, scalability, accuracy, etc.; • Introduction to the theoretical limitations of the computational methods; • Comparison of different computational methods. Part 3: Selection of software tools • Introduction to the selection of software for the computation methods;

- Performance benchmarking

Teaching/ Learning Methodology

Students need to select a particular class of computational problem, such as finite element, CAD or soft computing (artificial intelligence), and pursue an independent study of the subject. This subject comprises tutorials and seminars. Theories and concepts of different basic computational methods are conveyed by tutorials. Seminars are arranged to deepen students' knowledge on the advanced computational methods and theoretical consideration when applying the methods in real world problems. Group/individual projects and assignments are employed as the assessment tools for students' strategic thinking, team working and presentation skills.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				
		a	b	c	d	
1. Project(s)	45%	✓	✓	✓	✓	
2. Examination	55%	✓	✓	✓	✓	
Total	100%					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

- Group/individual projects and assignments are employed as the assessment tools for students' strategic thinking, team working and presentation skills.

- Examination is the integration of their understanding, ability to formulate and handle the problem.

Student Study Effort Expected

Class contact:	
▪ Tutorials	12 Hrs.
▪ Seminars	14 Hrs.
Other student study effort:	
▪ Reading	69 Hrs.
▪ Project	40 Hrs.
Total student study effort	135 Hrs.

Reading List and References	[Note: the reading list may change according to the topic that is chosen by the students. The following lists are examples.] <u>Topic: Soft Computing</u> 1. Soft computing in textile engineering [electronic resource]. Majumdar. Woodhead Publishing, 2011. 2. Kansei engineering and soft computing [electronic resource]: theory and practice / Ying Dai, Basabi Chakraborty, and Minghui Shi, editors. Hershey, Pa.: IGI Global, 2011. 3. Marketing intelligent systems using soft computing [electronic resource]: managerial and research applications / Jorge Casillas and Francisco J. Martínez- López (Eds.) Berlin: Springer-Verlag, c2010. 4. Artificial intelligence and soft computing [electronic resource]: 10th international conference, ICAISC 2010, Zakopane, Poland, June 13-17, 2010 / Leszek Rutkowski ... [et al.] (Eds.)ICAISC 2010 (2010: Zakopane, Poland) Berlin: Springer, c2010. <u>Topic: Optimization</u> 1. Supply chain optimization, management and integration [electronic resource]: emerging applications / John Wang, editor. Hershey, Pa.: IGI Global, c2011. 2. Optimization and optimal control [electronic resource]: theory and applications / edited by Altannar Chinchuluun ... [et al.]. New York: Springer Science+Business Media, LLC, c2010. 3. Engineering optimization [electronic resource]: theory and practice / Singiresu S. Rao. Hoboken, N.J.: John Wiley & Sons, c2009. 4th Ed. 4. An introduction to management science [electronic resource]: quantitative approaches to decision making / David R. Anderson, Dennis J. Sweeney, Thomas A. Williams. Cincinnati, Ohio: South-Western College Pub., 2000. 9th Ed. <u>Journals</u>*:* 1. Clothing And Textiles Research Journal 2. Computer Aided Design 3. International Journal Of Clothing Science And Technology 4. Journal of Textile And Apparel, Technology And Management 5. Journal of Textile Institute 6. Research Journal of Textile And Apparel 7. Sen'i Gakkaishi 8. Taiwan Textile Research Journal 9. Textile Research Journals